System Configuration for Deployment

Project Overview

Flask-based backend application providing OCR processing, face liveness verification, anti-spoofing detection, image processing, machine learning services, and Supabase-based authentication.

Recommended Production Server Configuration

* Operating System: Ubuntu 22.04 LTS
* CPU: 8 vCPU Cores

+ RAM: 16 GB

* Storage: 100 GB SSD

* Python: 3.11+

* Reverse Proxy: Nginx

* Application Server: Gunicorn

Database & Authentication

* Supabase (PostgreSQL Database)
* Supabase Authentication

+ JWT-based Authentication

* Secure API Access

Major Backend Dependencies

Flask 3.1.3, Flask-CORS 6.0.2, Flask-JWT-Extended 4.7.1, OpenCV 4.13.0, OpenCV-Contrib 4.13.0, MediaPipe 0.10.35, CVLib 0.2.7, CVZone 2.0.0, NumPy 2.4.4, SciPy 1.17.1, Scikit-Learn
1.8.0, Scikit-lmage 0.26.0, Matplotlib 3.11.0, Pillow 12.2.0, Requests 2.34.2 and supporting
libraries.

Deployment Architecture

React Frontend Flask Backend API OCR/Liveness / Anti-Spoofing Modules
React Frontend Flask Backend API Supabase Authentication & Database

Environment Variables

SUPABASE_URL

SUPABASE_ANON_KEY
SUPABASE_SERVICE_ROLE_KEY (if applicable)
JWT_SECRET_KEY